

Week 5: Database Design & Normalization

CST4714 Database Administration —
Thursday Session

Session Roadmap

- Warm-up: surface schema pain points from drafts
- Concept primer: ER modeling refresh
- Keys & dependencies in practice
- Normalization ladder demonstration
- Live rebuild demo of the simplified schema
- Design studio lab + share-out
- Midterm alignment & next steps

Learning Objectives

- Translate requirements into entities, relationships, and attributes
- Select key strategies that encode real business rules
- Diagnose redundancy using functional dependencies
- Apply 1NF $\rightarrow$ 3NF (and note when BCNF is realistic)
- Document trade-offs so the midterm proposal stays consistent

Why ER Modeling Matters

- Shared vocabulary between stakeholders and engineers
- Shapes how we scope tables, constraints, and permissions
- Flags missing business rules before code is written

Entities, Attributes, Relationships

- Entities = nouns we store (Student, Advisor, Plan)
- Attributes = descriptors tied to one entity
- Relationships = verbs/cardinality: one-to-many, many-to-many
- Optionality drives foreign key nullability

Keys & Constraints

- Natural vs surrogate keys: choose intentionally
- Composite keys capture multi-attribute uniqueness
- Alternate keys support lookups (email, account number)
- Constraints = truth serum: PRIMARY, UNIQUE, CHECK, FK

Functional Dependencies

- Express business rules: determinant $\rightarrow$ dependent
- Spot partial dependencies on part of a composite key
- Find transitive chains (Advisor $\rightarrow$ Email)
- Use dependency lists to drive decomposition

First Normal Form (1NF)

- Enforce atomic columns and consistent data types
- No repeating groups or multi-valued cells
- Primary key uniquely identifies each row

Second Normal Form (2NF)

- Remove attributes that depend on part of a composite key
- Factor shared facts into their own tables
- Example: plan metadata should not live on attendance rows

Third Normal Form (3NF) & BCNF

- 3NF: non-key attributes depend only on the key
- BCNF: every determinant is a candidate key
- Check for Advisor → Email or Session → Instructor chains
- When BCNF conflicts with practicality, document it

Design Trade-Offs

- Denormalize on purpose for performance/reporting
- Index choices follow from access patterns
- Use comments and diagrams to capture intent

Demo Build Checklist

- Load the raw spreadsheet data into public schema
- Run anomaly queries (capacities, multi-date orders)
- Create normalized tables with simple keys
- Insert distinct dimension rows + attendance facts
- Show a clean join result to prove the fix

Lab Reminders

- Run the setup SQL to recreate wellness_subscription_raw
- Map functional dependencies before touching DDL
- Create six core tables (students, advisors, plans, sessions, subscriptions, attendance)
- Populate tables with INSERT ... SELECT DISTINCT
- Capture one proof query and write a short reflection

Demo Roadmap

- Start with the messy ticketing sheet
- Surface anomalies with quick diagnostics
- Design the normalized tables with students/advisors/plans/sessions
- Insert distinct values and link attendance records
- Run one clean query to confirm the design

Lab Expectations

- Run the setup script to rebuild the raw table
- List functional dependencies before writing DDL
- Create six core tables with primary/foreign keys
- Populate tables using `INSERT ... SELECT DISTINCT`
- Run one proof query to confirm the design
- Write a short reflection on remaining trade-offs

Schema Review Roundtable

- Teams share one big change and its justification
- Class critiques edge cases and denormalization
- Identify open questions for office hours

Midterm Alignment

- Update ERDs with new entities/relationships
- Map business rules to constraints in proposals
- Flag data quality assumptions for stakeholders

Takeaways & Next Steps

- Normalize first, denormalize with intention
- Constraints are living documentation
- Prep questions for next office hours session
- Start integrating feedback into midterm drafts